

Beam Transmission and Extinction in the Mu2e Beam Line

Eric Prebys

May 7, 2014

Abstract

A g4beamline simulation has been performed for the section of the Mu2e beam line from the AC dipole to the production target. Beam transmission was calculated as a function of deflection in the AC Dipole, using a *single* collimator placed 90° in phase advance downstream. Results were found to be in good agreement with first principle calculations. The resulting transmission function was then used to re-optimize the AC dipole harmonic content, based on recent simulations of the longitudinal bunch shape from the Debuncher. It was found that a transmission probability of $< 5 \times 10^{-8}$ or less for out of time beam can be achieved with this configuration, while maintaining a transmission efficiency of well over 99% for in time beam. Achieving this high level of extinction requires that the beam does not hit anything upstream of the first collimator. This will necessitate either increasing the aperture in the region of quads 924, 925, and 926 or adding a third (900 kHz) harmonic to reduce maximum amplitude of AC dipole system. Both options are discussed.

1 Principle of Operation

Elsewhere[1], we have discussed in detail the relationship between requirements of an AC dipole and the local beam line parameters. The effect of the extinction magnet/collimator system can be represented as a translation in transverse phase space, as illustrated in Figure 1. We can discuss the effect of the AC dipole in a manner that is independent of the local optics if we normalize the bend angle to the angle corresponding to the boundary of the admittance; that is

$$\theta_0 \equiv 2\sqrt{\frac{A}{\beta_D\beta\gamma}} \quad (1)$$

where β_D is the betatron function at the location of the AC dipole (assuming it is an extremum). We assume that a collimator is placed 90° away in phase advance, and that the half gap d is also set to the limiting admittance

$$d = \sqrt{\frac{A\beta_C}{\beta\gamma}} \quad (2)$$

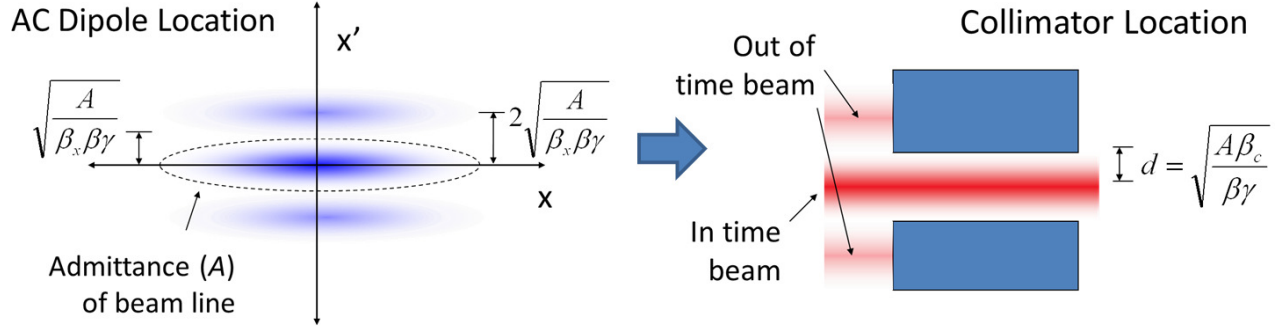


Figure 1: Effect of the extinction magnet in phase space. Beam line admittance A is indicated by the ellipse. Shown at right is effect of the dipole at the collimator (or other defining aperture).

We can then define a normalized deflection angle δ as

$$\delta \equiv \theta/\theta_0 \quad (3)$$

Thus, $\delta = 1$ will put the center of the beam just at the edge of the collimator, resulting in a 50% transmission. An angle of $\delta = 2$ will result in all beam within the allowed admittance striking the collimator. We therefore refer to $\delta = 2$ as the “extinction angle”. An ideal “black hole” collimator would have exactly zero transmission beyond this angle. The goal of this study is to determine the actual transmission in the Mu2e beam line with a realistic collimator.

The baseline system which has been discussed up until now is based on two harmonic components, provided by six identical 1-meter magnets in two groups of three. The low frequency component is 300 kHz, phased such that the nominal Mu2e 600 kHz bunches occur at the nodes, while out of time beam will be swept into a collimator. The high frequency (3.8-5.1 MHz) component is designed to reduce the slewing during the transmission window to maximize transmission efficiency. The amplitude of the low frequency component of the AC dipole is set to give a value of $\delta = 2$ at the edges of the transmission window. This means that the maximum amplitude will be much larger than that required to extinguish the out of time beam. As we will see in Section 4, the combination of the two components will have an amplitude of about $\delta = 10$ in the case of the optimized transmission window. This leads to some problems with upstream scattering which must be addressed.

2 Beam Line Design

The optics of the current beam line design are shown in Figure 2[2]. As indicated in the figure, there are five potential locations for collimators. The second, at location 928, is located at betatron phase advance of 88.9° from from the AC dipole in the horizontal (bend) plane, making it the ideal

location for the extinction collimation. *This is the only collimator we will consider in our study.* The betatron functions at the two locations are shown in Table 1.

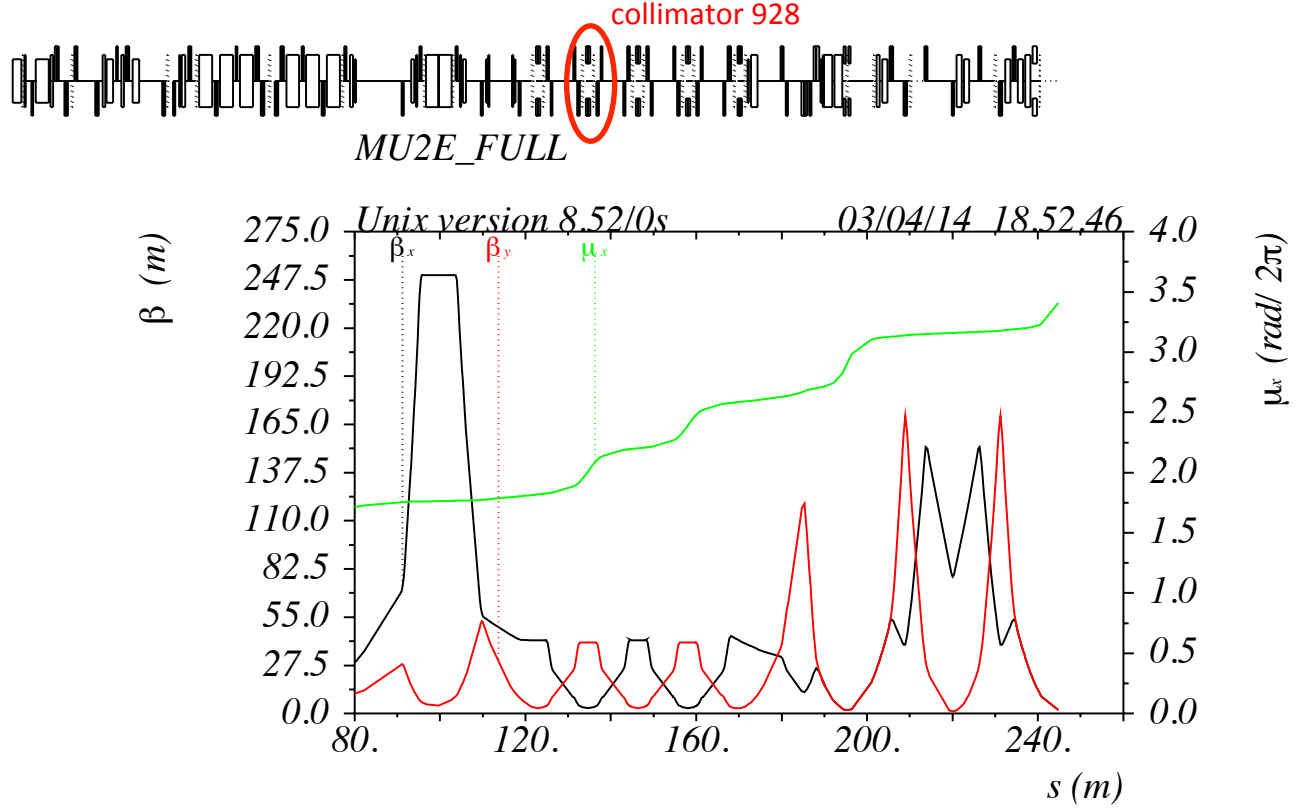


Figure 2: Beam optics for the Mu2e beam line, plotted from the AC dipole to the production target. Potential collimator locations are indicated in the beam line schematic at the top. Collimator 928 is located 89° from the AC dipole in phase advance in the bend plane.

Table 1: Betatron functions at the AC dipole and extinction collimator locations.

Location	β_x [m]	β_y [m]
AC Dipole	250.18	4.50
Collimator	3.15	40.57

3 G4beamline Simulation

A PERL script was used to convert the MAD optics file to a g4beamline script[3]. Only the portion of the beam line from the AC Dipole to the target was considered. This part of the beam line

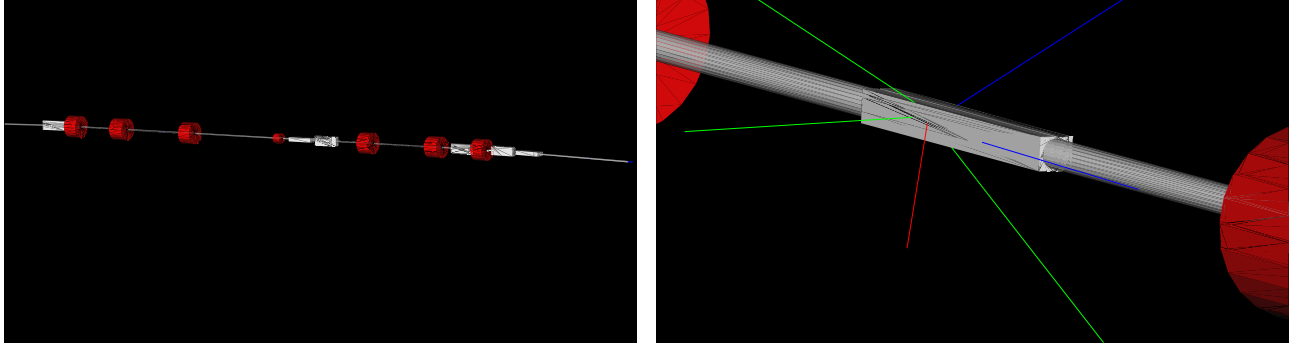


Figure 3: Sections of the G4beamline model of the Mu2e beam line. On the left is shown the end of the line, including all three types of magnets. The right shows an out of time particle hitting the collimator.

Table 2: Supplied and calculated beam parameters.

Parameter	Value	Description
K	8 GeV	Beam kinetic energy
$\epsilon_{x,\text{full}}$	$30 \pi\text{-mm-mrad}$	Normalized full uniform emittance in x plane
$\epsilon_{y,95}$	$15 \pi\text{-mm-mrad}$	Normalized 95% Gaussian emittance in y plane
A	$50 \pi\text{-mm-mrad}$	Full normalized beam admittance
x_{max}	28.15 mm	Maximum of x distribution at center of AC dipole
x'_{max}	1.13×10^{-4}	Maximum of x' distribution at center of AC dipole
σ_y	1.09 mm	RMS of y distribution at center of AC dipole
$\sigma_{y'}$	2.42×10^{-4}	RMS of y' distribution at center of AC dipole
d	4.08 mm	Half-gap of collimator.
θ_0	.1452 mrad	Angular deflection corresponding to admittance (Eq. 1)
$(BL)_0$	43.07 T-m	Integrated field corresponding to θ_0

consists of three types of magnets: “SQ” and “LQ” quadrupoles of various lengths, as well as “CR” bend dipoles. The g4beamline descriptions of these magnets are based on Technical Division drawings. There are two vertical CR magnets which are integral to the design, while four additional CR magnets are designed to be used for target scanning and have their fields set to zero for this study. The collimator consists of two stainless steel jaws which are 10 cm high (y) by 5 cm wide (x) by 1 m long (s). The remaining space is filled with a 3.5” OD stainless steel beam pipe with a 1.5 mm wall. Trim magnets, toroids, BPMs, stands, etc are omitted. Sections of the beam line model are shown in Figure 3. Note that because this line contains only vertical bends, it can be assumed to be left/right symmetric with respect to the AC dipole deflection.

We do not yet have a detailed simulation of the beam extracted from the Delivery ring, so our model here is based on very conservative assumptions[4]. Beam is extracted in the horizontal

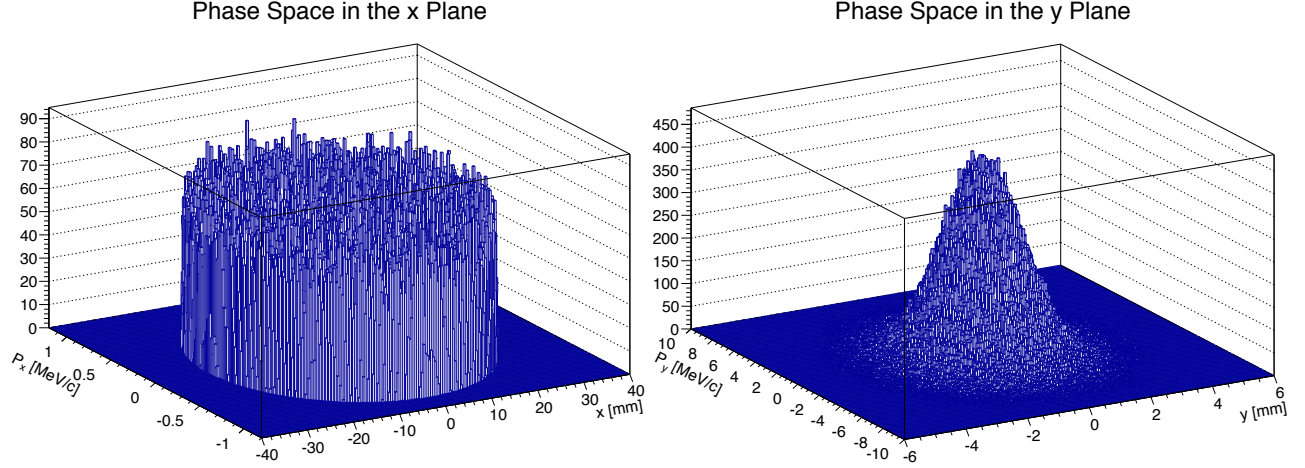


Figure 4: Initial phase space distributions in the two planes.

plane, which is also the bend plane of the AC dipole. The extraction channel corresponds to roughly 10π -mm-mrad normalized emittance. Allowing for some mismatch, we assume 100% of the beam will be contained within a normalized emittance of 30π -mm-mrad. In this simulation, this is modeled as a uniform distributions in two dimensional phase space. To first order, the phase space in the vertical plane should not be affected by extraction, so it will retain the emittance in the Delivery ring, which should be reasonably represented by a Gaussian beam with a 95% normalized emittance of 15π -mm-mrad. The collimator jaw gap is set corresponding to a 50π -mm-mrad normalized admittance, which corresponds roughly to the design admittance of the AC dipole itself. All of the key parameters are shown in Table 2, and the initial particle distributions are shown in Figure 4.

The QGSP_BERT physics model was used, with a kinetic energy cut of 100 MeV, in order to speed up computation while insuring that no surviving particle could generate background.

To verify the optical model, 10000 protons were generated and the lattice functions calculated using the G4beamline “profile” command. The results are shown in Figure 5, which shows good agreement with the MAD optics (note: for this study, Gaussian distributions were used in both planes).

To study transmission, sets of particles were generated with values of the magnetic field in the AC dipole corresponding to particular values of the normalized bend angle δ , in increments of .1.

During the studies, it was found that for bend angles greater than $\delta=5.4$, some particles will scatter off of the beam pipe and or magnets upstream of the primary collimator, resulting in beam scattered back into the transmission channel, as illustrated in Figure 6.

To prevent these problems, we attempted to use collimator 924, centered at $s=123.453$ m to capture the high angle beam before it hit the beam pipe wall; however, it was found that again, the particles near the edge of upstream collimator were scattered back into the beam and hit the

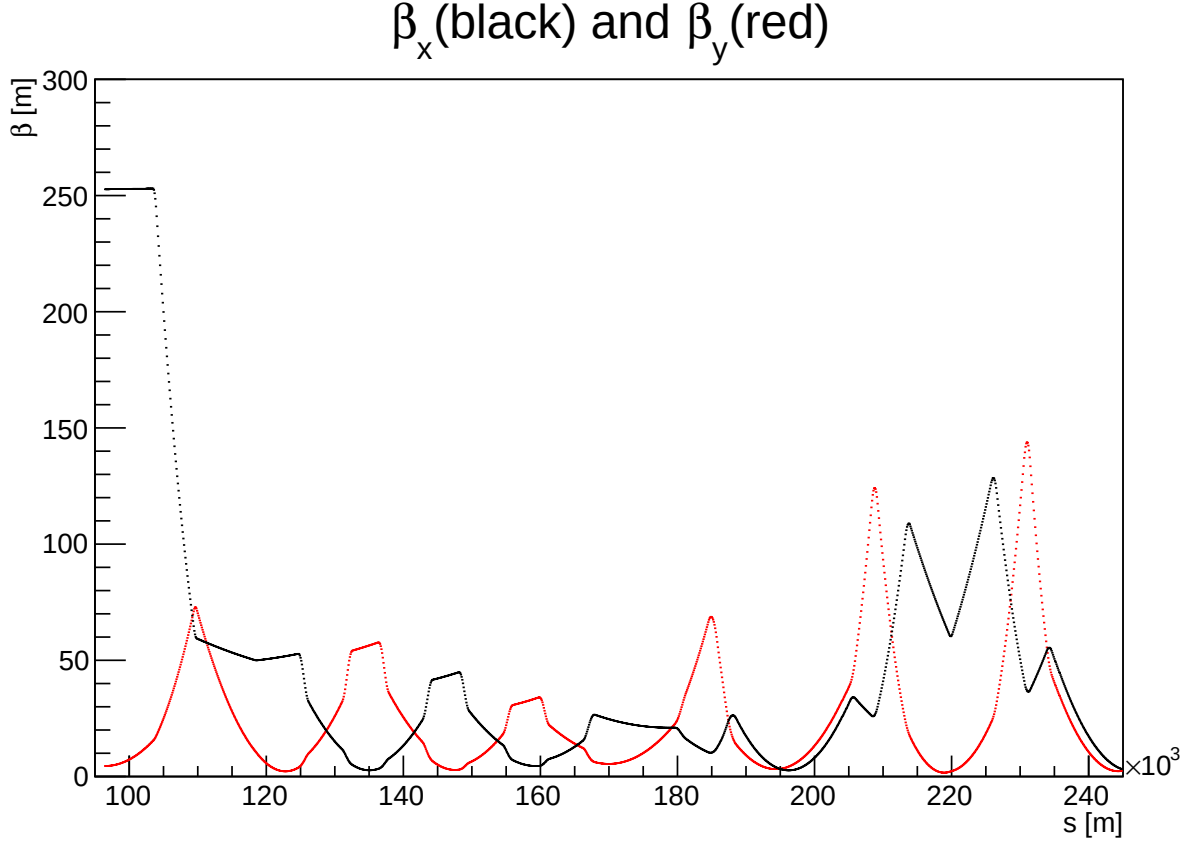


Figure 5: Fitted optics for the g4beamline model. These are in reasonable agreement with the MAD calculation.

target at the 10^{-6} level. The addition of a third and fourth collimator downstream would reduce the transmission somewhat, but the overall performance never again achieved the 10^{-8} level. Although the integrated extinction was still within the specifications, this degraded performance was viewed as undesirable.

For this reason, we removed the 924 collimator and replaced quads 924, 925, and 926 with LQ quads, as illustrated in Figure 7. We also increased the beam pipe to 6" from quad 923 to 927. This increased aperture means that even at the highest deflection angles, there are no beam hits upstream of the 928 collimator. Simulations were re-run for this new configuration for the high angle deflections, and these are the results which will be shown here. Another solution would be to add a third harmonic to reduce the maximum deflection, and this will be discussed in Section 5.

The diameter of the target will be 6.3 mm, but to compensate for any potential misalignments or problems caused by incorrect modeling, we conservatively count a “hit” in the extinction sense

Table 3: Table of number of generated protons and number which fall within a 5 mm radius at the target location.

δ	Generated	Survived
0.0	1000	1000
0.1	1000	1000
0.2	5000	5000
0.3	5000	4930
0.4	5000	4688
0.5	5000	4401
0.6	5000	4027
0.7	5000	3703
0.8	5000	3323
0.9	5000	2944
1.0	5000	2500
1.1	5000	2110
1.2	5000	1733
1.3	5000	1361
1.4	10000	1999
1.5	10000	1359
1.6	50000	4016
1.7	100000	3544
1.8	1000000000	604260
1.9	1000000000	266
2.0	1000000000	43
2.1	1000000000	16
2.2	1000000000	11
2.3	1000000000	4
2.4	1000000000	5
2.5	1000000000	5
3.0	1000000000	1
4.0	1000000000	5
5.0	1000000000	2
6.0	1000000000	5
7.0	1000000000	1
8.0	1000000000	5
9.0	1000000000	3
10.0	1000000000	4

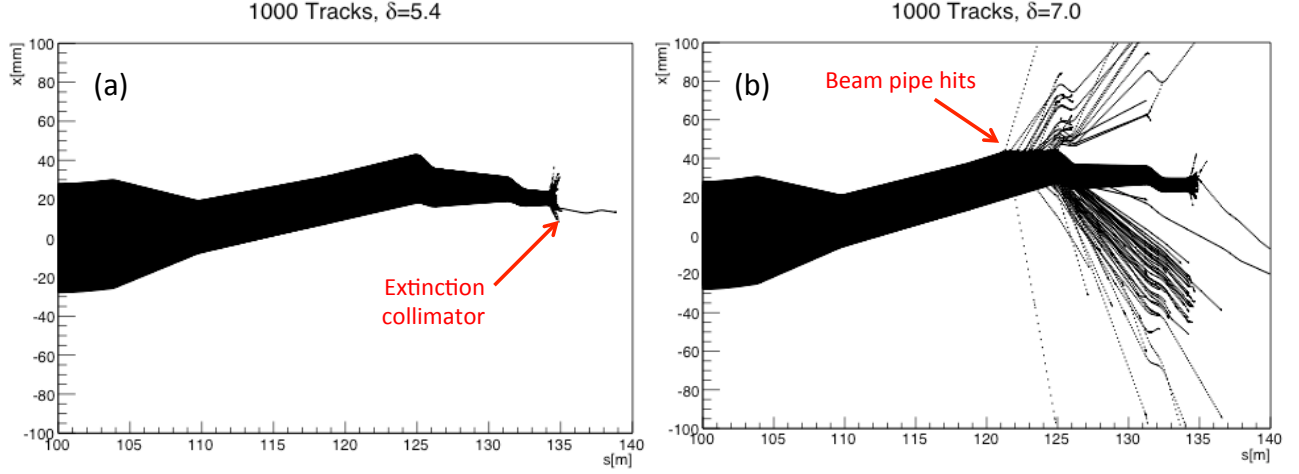


Figure 6: Scattering from beam pipe. Figure (a) shows the first 1000 tracks for $\delta = 5.4$, none of which hit the beam pipe. For values of δ beyond this, some particles hit the beam pipe and scatter back into the beam line, as shown in (b).

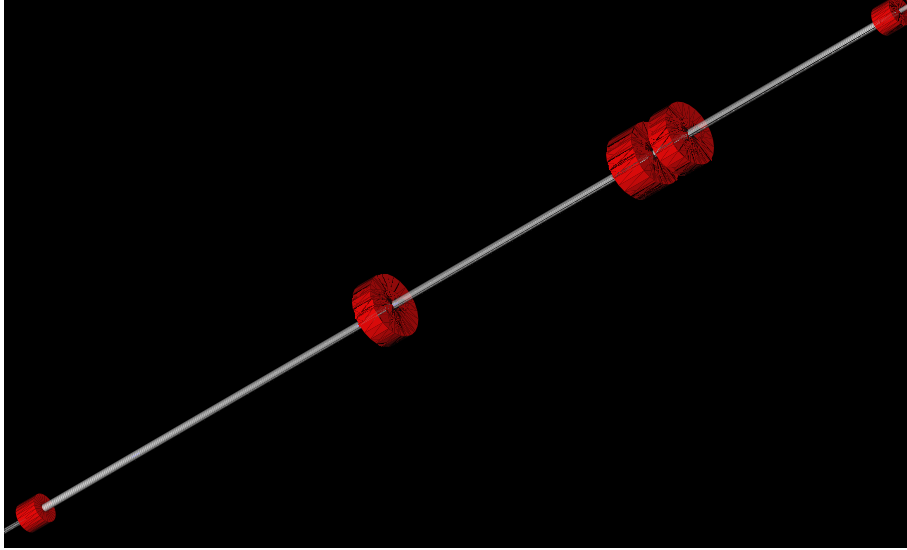


Figure 7: In order to eliminate upstream scattering, quads 924, 925, and 926 have been replaced with LQ quads. The beam pipe has also been replaced with 6" beam pipe from Q923 to Q927.

to be any particle which falls within a 5 mm radius at the location of the target¹ In each case,

¹The phase advance from the AC dipole to the target insures that *any* particle which passes through the collimator will hit the target, so this cut will not lead to an overestimate of the transmission efficiency.

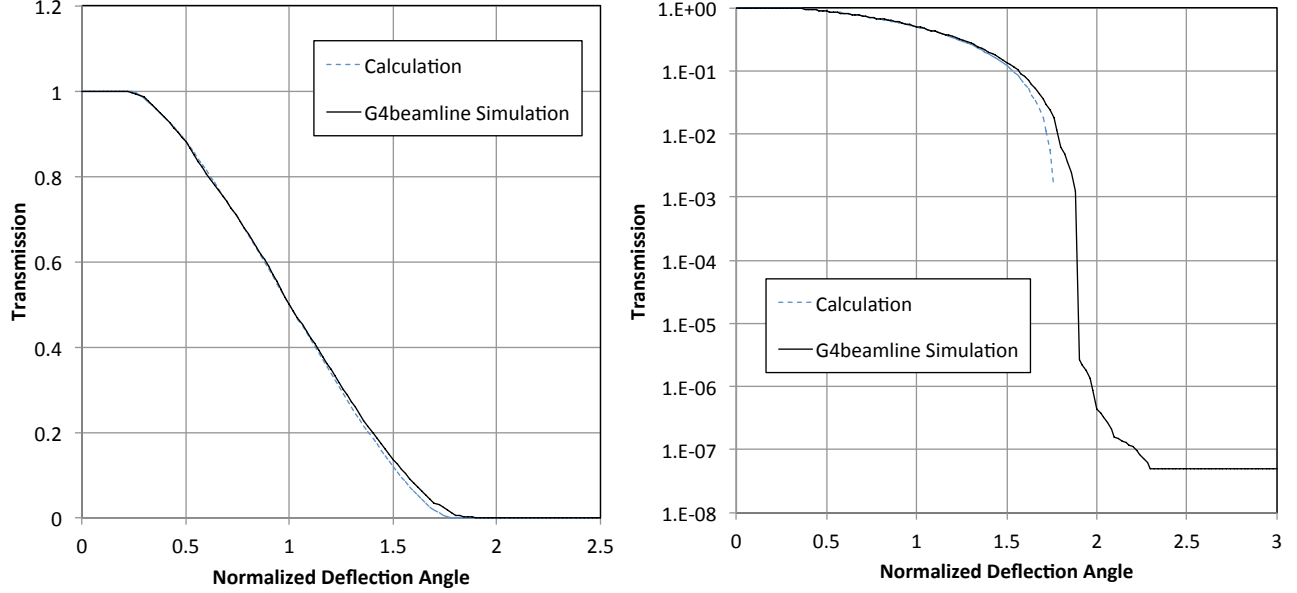


Figure 8: Transmission as a function of normalized deflection angle for a beam described in the text. The solid line shows a full g4beamline simulation while the dotted line shows the calculation assuming an ideal collimator and a mathematical description of the beam. As indicated in the logarithmic plot at right, samples with 5 or less surviving particles out of 10^8 generated protons were assigned a transmission of 5×10^{-8} .

enough initial particles were run so that at least 1000 survived, up to a maximum of 10^8 . Sets with more than a million initial particles were run on the grid. Results are shown in Table 3. We see that for values of δ of 2.4 and above, five or fewer particles survived out of 10^8 , so the transmission is uniformly set to 5×10^{-8} for this range.

Transmission is plotted in Figure 8. Also shown is the curve one would obtain from a simple calculation assuming a perfect collimator.

4 Optimization of AC Dipole Harmonics

The harmonic content of the AC dipole system has been optimized before. The general procedure is to choose an odd harmonic of the fundamental waveform, define the first nodes of this harmonic as the boundaries of the transmission window and normalize the 300 kHz harmonic to produce full extinction (i.e. provide a normalized deflection angle of $\delta = 2$) at the edges of this window. This procedure is repeated with three significant differences:

- The original study was done using a 95% normalized Gaussian emittance of 20π -mm-mrad. This has been replaced with a 100% normalized uniform emittance of 30π -mm-mrad, as

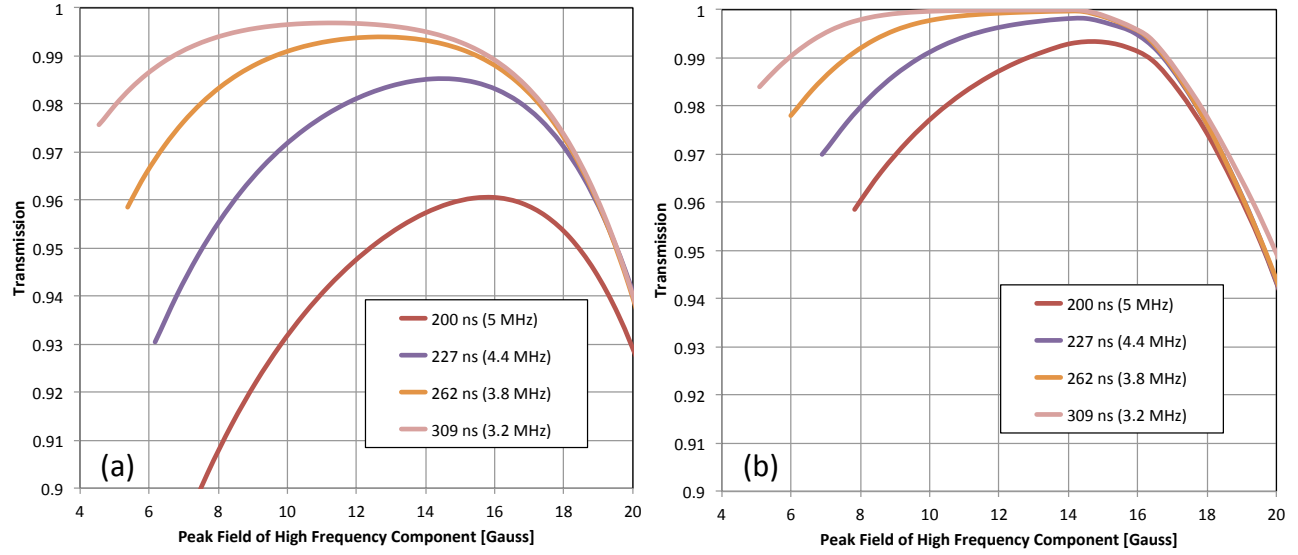


Figure 9: Transmission efficiency as a function of higher harmonic amplitude, assuming 3 1-meter high frequency magnets. Plot (a) shows the previous study, while plot (a) shows the current study with new bunch distribution, more correct emittance, and full simulation for the transmission function.

Table 4: Harmonic content of the AC dipole, shown for both the two component baseline scheme and the proposed three harmonic scheme.

Configuration	Harmonic Amplitude [G-m]			Peak Amplitude (normalized)
	300 kHz	900 kHz	4.5 MHz	
Two Harmonic (baseline)	413.7	-	35.2	10.4
Three Harmonic (proposed)	206.7	73.2	35.2	5.4

discussed in Section 3.[4].

- The longitudinal distribution has been replaced with one generated by a more up-to-date simulation, which has much smaller tails than the previous simulation[5].
- The transmission is based on the simulation shown in Figure 8, rather than a naive calculation, although in terms of transmission efficiency, the two are almost identical.

The improved longitudinal distribution has resulted in a dramatic improvement in performance, as illustrated in Figure 9. Figure 9(a) shows the previous study, which drove us to choose the 13th harmonic, resulting in a widened transmission window. In contrast, 9(b) shows the new study. We see that we now get very good transmission with the 15th harmonic, allowing the full transmission window to be reduced to 227 ns. With the specified field of the high frequency dipole of 12 Gauss in

3 1-meter magnets, we get a transmission efficiency of approximately 99.7%. The optimized values of the integrated fields are shown in the first row of Table 4.

5 Addition of Third Harmonic

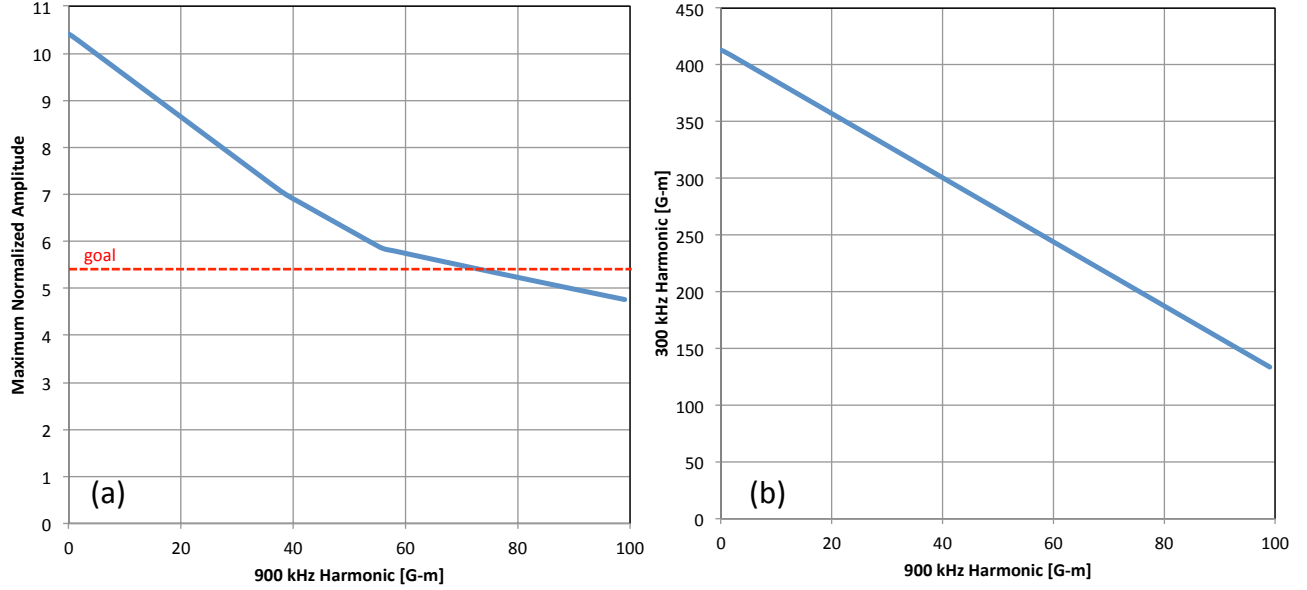


Figure 10: The dependence of the maximum normalized amplitude on the peak integrated field in the third (900 kHz) harmonic is shown in (a). Peak amplitude is shown as the normalized angle $\delta \equiv \theta/\theta_0$, where θ_0 is the angle required to move the center of the beam to the edge of the collimator. In these units, the maximum amplitude must be kept at or below $\delta = 5.4$ to avoid unwanted collisions. The dependence of the required integrated field of the first (300 kHz) harmonic on the third harmonic is shown in (b).

As stated earlier, it was found that this high amplitude resulted in some beam hitting the beam pipe and/or magnets upstream of the primary collimator, resulting in some beam scattering back into the transmission channel. To complete the study, the upstream aperture was enlarged to prevent this, but this may not prove practical in real life. Another solution is to add a third (900 kHz) harmonic to reduce the maximum amplitude, which we investigate in this section.

It was important to keep the waveform the same during the transmission window, so the sum of the 300 kHz and 900 kHz waveforms were constrained to have a normalized amplitude of $\delta = 2$ (full extinction) at ± 114 ns. This means that adding the third harmonic reduces the required amplitude of the first harmonic. The amplitude of the 15th harmonic (4.5 MHz) component, which has nodes at ± 114 ns, was kept at the optimum value determined in Section 4.

Figure 10(a) shows the dependence of the maximum normalized deflection amplitude on the integrated field in the third harmonic. We see that an integrated field of 73.2 G-m will keep the maximum amplitude below $\delta = 5.4$, as required to prevent collisions. This will also lower the required integrated field in the first harmonic to about 206.7 G-m, as shown in 10(b). The final values are given in the second row of Table 4.

The AC dipole system is composed of 6 identical 1-meter segments. The baseline configuration has 3 used for the 300 kHz component and 3 used for the 4.5 Mhz component, resulting peak fields of 138 and 12 Gauss, respectively, based on the integrated fields in the first row of Table 4.

The new proposal would be to continue using three magnets for the 4.5 MHz components, but to re-task one of the remaining magnets to the 900 MHz component, with a peak field of 73.2 Gauss. The remaining two magnets would operate at 300 kHz, with a peak field of 104 Gauss, i.e. significantly *lower* than the baseline scheme.

It is not believed that the 300 kHz power supply can be adapted to such a high frequency[6], so this component would be powered by an RF amplifier, similar to the 4.5 MHz compoent. We are working out the power requirements, but based on simple interpretation, we believe it will be less than 2 kW[7], putting it in the range of the spare supplies we are planning to use for the high frequency component.

6 Final Waveform and Transmission Window

Figure 11 shows the optimized waveform for both the two and the three harmonic cases. We see that as specified, it hits the extinction amplitude of $\delta = 2$ at ± 114 ns.

Figure 12 shows the transmission window, along with the old and new simulated bunch distributions. We see that the old bunch distribution would have had an unacceptable efficiency with this transmission window. The final waveforms and transmission curves are stored along with this document as

“final_transmission.cvs” and “final_transmission.xlsx”. The transmission is $\leq 5 \times 10^{-8}$ for all times outside of the nominal transmission window of ± 125 ns. When combined with the expected performance of the Delivery Ring[5], this more than satisfies the extinction requirement for the experiment[8].

7 Target Position Slewing

Although the high frequency harmonic has been added to reduce slewing at the nominal transmission time, there is still some possility that the time-slewing will result in a widening of the effective distribution at the target. To investigate this, a numerical integration was done to convolute the position slewing caused by the AC dipole with the time distribution and transmission curve based on the most recent simulations, assuming the beam distribution described in Section 3. The results

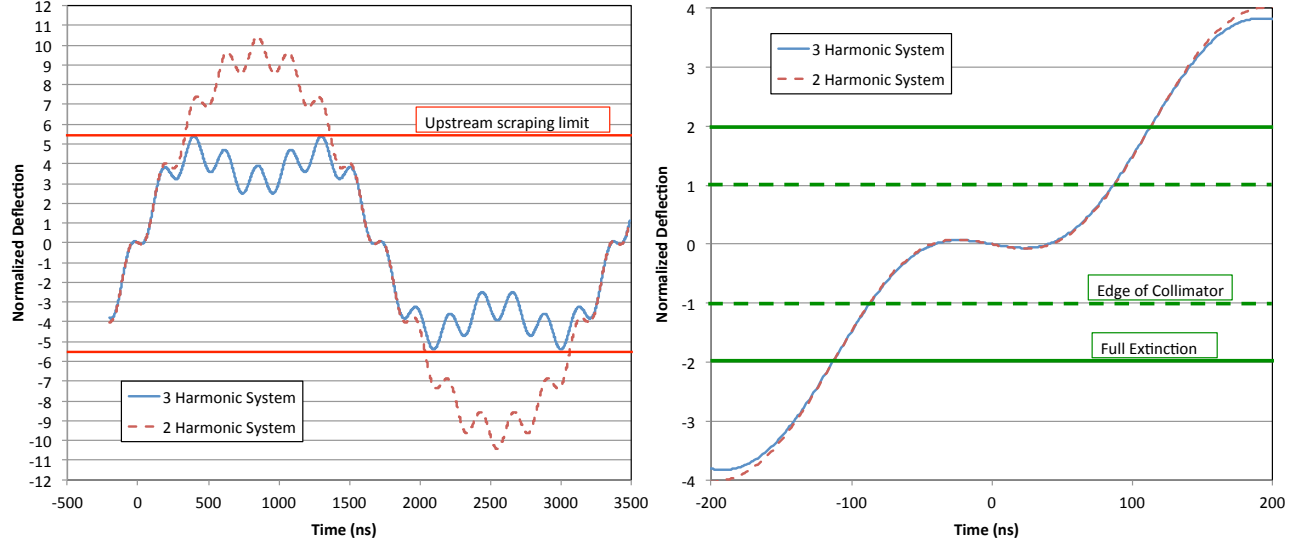


Figure 11: Optimized waveform. The dotted line shows the two harmonic waveform, while the solid line shows the addition of a 900 kHz harmonic to reduce the maximum amplitude and avoid upstream scattering. The plot on the left shows the waveform over one cycle. The one on the right shows a closeup of the waveform during the transmission window.

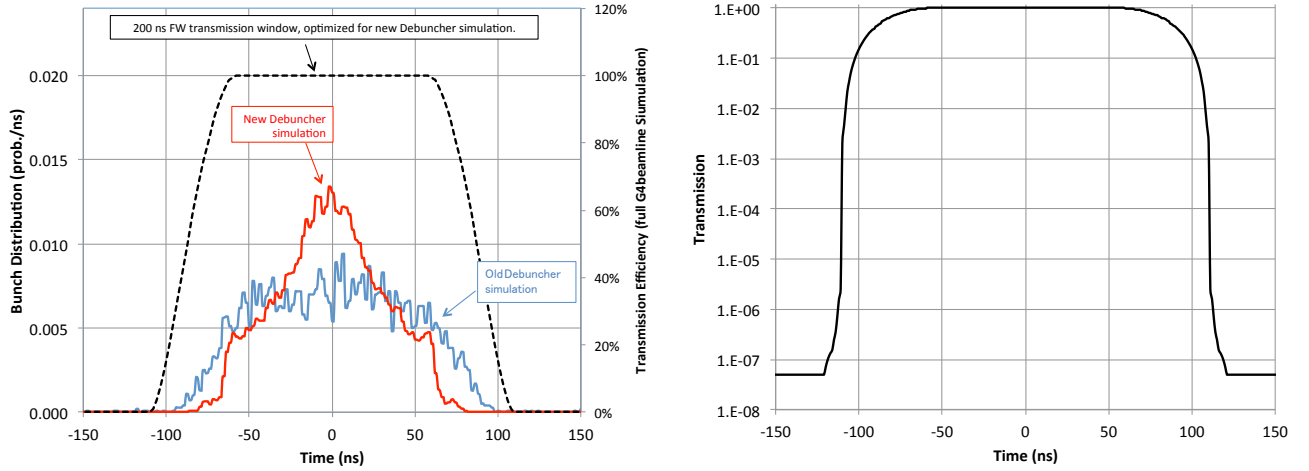


Figure 12: The plot on the left shows the fully simulated transmission window, with the new and old bunch distributions shown. The plot on the right shows the transmission window on a logarithmic scale. As stated previously, large amplitude deflections were conservatively assigned a transmission of 5×10^{-8} .

are shown in Figure 13. As can be seen, the optimized waveform results in negligible slewing relative

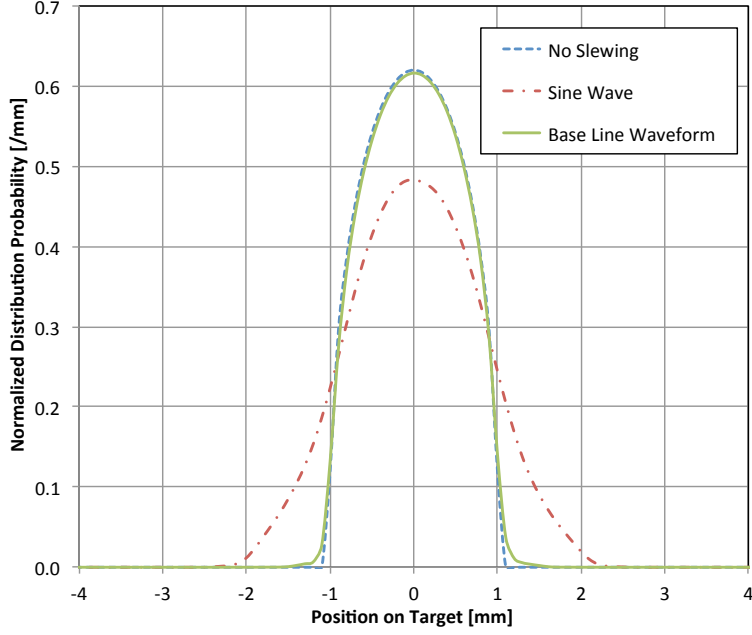


Figure 13: Normalized distribution at the target, comparing the optimized waveform to the case of no slewing at all. Also shown is the distribution one would get using a single-harmonic sine wave, which would slew linearly over the transmission window.

to the case with no deflection at all. For comparison, the effect of a single harmonic dipole is shown, which would result in linear slewing over the transmission window. In this case, slewing would be significant.

8 Beam Admittance Issues

This analysis relied critically on the assumption that there would be no beam outside the 30π -mm-mrad admittance ellipse that could be transmitted at unacceptable levels. It's possible that upstream scatters could give rise to such beam. Determining this will require simulation of the beam from the extraction to the position of the AC dipole, and then calculating the fraction of that beam that will fall into the transmission channel. For the time being, it is recommended that we reserve two collimators to define the horizontal admittance upstream of the AC dipole, if required.

9 Conclusions

A full G4beamline simulation has been carried out, using the final design of the Mu2e beam line for the region from the AC dipole to the target. It was found that to achieve the required performance,

it was necessary to avoid any collisions upstream of the primary extinction collimator. This can be done either by increasing the aperture of the beam line through quadrupoles 924, 925, and 926, or by limiting the amplitude of the AC dipole system. Both options are being considered, and have been discussed in this note.

After this correction was made, harmonic components were optimized using recent simulations of the longitudinal bunch shape, and it was shown that a transmission well over 99% can be maintained for the in time particles, while an extinction level of $\leq 5 \times 10^{-8}$ or less can be achieved for particles more than 125 ns from the nominal bunch time, using a single, 1 m long stainless steel collimator. This satisfies the requirements of the Mu2e experiment.

References

- [1] E. Prebys, “Extinction Magnet Specifications for the Mu2e Experiment”, <http://mu2e-docdb.fnal.gov>, Mu2-doc-709
- [2] C. Johnstone, *private communication*.
- [3] Script was written and run by Jean-Francois Ostiguy.
- [4] V. Nagaslaev, *private communication*.
- [5] S. Werkema, *private communication*.
- [6] H.Pfeffer, *private communication*
- [7] V.Kashikhin, *private communication*
- [8] J. Miller, “Beam Extinction Requirement for Mu2e”, <http://mu2e-docdb.fnal.gov>, Mu2-doc-1175